RACE CAR | POWER SYSTEMS ONLY

INDIVIDUAL PHYSICAL CONDUCTORS / REV J / 25 OCT 2026

43 native nets | 85 field conductors | 41 supplier paths | 2 intact cable assemblies | 23 explicit joint / ground nodes

PROPOSED CONNECTION TOPOLOGY - unmeasured lengths, wire sizes, colors, terminals and splice hardware. NOT A FABRICATION RELEASE.

C / FIELD PATH
S / SUPPLIED LOOM
V / VENDOR CABLE
I / INTERNAL
B / BOND / METAL
U / UNRESOLVED

BUILD TAGS: MAKE = provide branch | LEAD = supplied wire, finish ends | LOOM = existing path | CABLE = intact cable | HOLD = verify

Black bars inside boxes are explicit splices/common metal/shared terminals. Crossings with white gaps never join.

Find by wire ID, net ID, component, joint or PDF bookmark. Click a component header for the complete native sheet.

VECTOR / SEARCHABLE | 119.9 x 46.2 inches

Line geometry is illustrative; vehicle routes and cut lengths are unmeasured.

02 / BATTERY, ISOLATION AND STARTER
BT1 Battery BRAILLE G30H
G1 Alternator 79376G / ONE-WIRE
F0 MAIN BUS FUSE / SELECT
F3 ALT BRANCH FUSE / AT STARTER
F1 Engine controller supply fuse R3 FEED / SELECT
U3 Cole Hersee 24059 starter relay COLE HERSEE 24059 / 12V / 85A
D60 Starter-relay coil suppression diode 1N5408 / BAND TO CMD
M1 Starter motor VW HIGH-TORQUE STARTER
J1 Engine controller power connector R3 MAIN SURLOK
F5 START DRIVER FUSE / SELECT
F2 Shared starter / charging cable fuse SHARED CABLE FUSE / AT BATTERY
J2 Engine ground strap GROUND STRAP / ENGINE
J3 Chassis ground strap GROUND STRAP / CHASSIS
ASSEMBLY A-J2 / body joins ends
ASSEMBLY A-J3 / body joins ends
18 / CARTEK GT, STOP SWITCH AND POSITIVE BLOCKS
U2 CARTEK GT battery isolator CARTEK GT / VERIFY s OR x
F4 GT CONTROL / 5A PROPOSED
TB1 Sierra FS46540 / PROPOSED FS46540 / RAW + / PROPOSED
TB2 Summit SUM-900533 / PROPOSED SUM-900533 / FUSED + / PROPOSED
S40 BAOMAIN latching stop / NC B00NTT91Y0 / LATCHING NC
03 / NEXUS R3 PIN ALLOCATION
J10 Engine controller - connector A R3 A / AMP34 KEY1
IGN SWITCH INPUT
HBO5 / START CMD
HBO1 / DBW MOTOR A
HBO2 / DBW MOTOR B
HBO3 / ABS IGNITION
HBO4 / COCKPIT 12V
HBO6 / PD16 WAKE
04 / ENGINE POWER, INJECTION AND IGNITION
J12 Haltech R3 - integrated PDM outputs R3 E / DTP4
HCO1 / INJECTORS
HCO2 / COILS
HCO3 / DW200
HCO4 / DW100 LIFT 1
INJ1 LS3 injector - cylinder 1 INJECTOR / CYL 1
IC1 Stock L94 coil - cylinder 1 COIL / CYL 1
INJ2 LS3 injector - cylinder 2 INJECTOR / CYL 2
IC2 Stock L94 coil - cylinder 2 COIL / CYL 2
INJ3 LS3 injector - cylinder 3 INJECTOR / CYL 3
IC3 Stock L94 coil - cylinder 3 COIL / CYL 3
INJ4 LS3 injector - cylinder 4 INJECTOR / CYL 4
IC4 Stock L94 coil - cylinder 4 COIL / CYL 4
INJ5 LS3 injector - cylinder 5 INJECTOR / CYL 5
IC5 Stock L94 coil - cylinder 5 COIL / CYL 5
INJ6 LS3 injector - cylinder 6 INJECTOR / CYL 6
IC6 Stock L94 coil - cylinder 6 COIL / CYL 6
INJ7 LS3 injector - cylinder 7 INJECTOR / CYL 7
IC7 Stock L94 coil - cylinder 7 COIL / CYL 7
INJ8 LS3 injector - cylinder 8 INJECTOR / CYL 8
IC8 Stock L94 coil - cylinder 8 COIL / CYL 8
SUPPLY
POWER RETURN
19 / STOCK L94 COIL LOOMS AND LS3 INJECTORS
X111 Stock L94 odd-bank coil connector OEM L94 / ODD / 7-WAY
X113 Stock L94 even-bank coil connector OEM L94 / EVEN / 7-WAY
SHARED COIL POWER RETURN
SHARED COIL +12V
EXPLICIT SPLICES / PROPOSED GROUND POINTS / SUPPLIER JOINS
SP103 ABS WAKE splice
S018 to J20/28
S019 to F43/2
S020 to H20/+
SP117 TAIL POS splice
SP121 CAMERA POS splice
OEM111-PWR Stock odd coil PWR splice
OEM113-PWR Stock even coil PWR splice
OEM-INJ-ODD Stock odd injector feed splice
OEM-INJ-EVEN Stock even injector feed splice
SUPPLIER-INTERNAL JOIN · actual construction/location UNCONFIRMED
Explicit join / no extra terminal numbers
Native sheet 02 / click header
J20 Anti-lock brake unit - main connector DELTA HCU / 38 PIN
F41 Anti-lock brake pump fuse ABS PUMP / VALUE TBD
F42 Anti-lock brake control fuse ABS MAIN / VALUE TBD
F43 Anti-lock brake wake supply fuse ABS WAKE / VALUE TBD
J21 Anti-lock brake ground connection ABS GROUND / HOLD
ASSEMBLY A-J21 / body joins ends
08 / DELTA ABS GEN 2 MAIN CONNECTOR
J22 Anti-lock brake vehicle interface PUBLIC DTM06-12S
S21 Anti-lock brake enable switch ABS ENABLE / GROUND ON
H20 Anti-lock brake warning lamp ABS WARNING LAMP
09 / ABS SENSORS, CONTROLS AND PUBLIC CAN
J41 Haltech PD16 - 25 A power outputs PD16 B / DTP4
U20 Haltech PD16 - main power PD16 / HT-198100
F50 PD16 main supply fuse AUX MAIN / TBD
H30 Left LED headlight LEFT HEADLIGHT / TBD
H37 Right LED headlight RIGHT HEADLIGHT / TBD
H31 Left tail LED bulb LEFT TAIL LED / TBD
H35 Right tail LED bulb RIGHT TAIL LED / TBD
H34 Left brake LED bulb LEFT BRAKE LED / TBD
H36 Right brake LED bulb RIGHT BRAKE LED / TBD
U22 Brake-marker electronics BRAKE MARKER / CONVERTER
IN
OUT
HEAD / HAZARD POWER
RETURN
TAIL / HAZARD POWER
BULB RETURN
BRAKE POWER
CAR SUPPLY IN
11 / AUXILIARY DISTRIBUTION AND LIGHTING
M3 Fuel lift pump 1 DW100 / LIFT 1
M4 Fuel lift pump 2 DW100 / LIFT 2
M5 High-pressure fuel pump DW200 / HIGH PRESSURE
M6 Radiator fan 1 PROFORM 67048 / 10 A
M7 Radiator fan 2 PROFORM 67048 / 10 A
PUMP POSITIVE
SWITCHED +12V
07 / FUEL PUMPS, SENDERS AND COOLING
S1 Engine controller power switch NORMAL R3 WAKE / U2 PERMIT
U10 Haltech iC-7 digital dash HALTECH IC-7
U11 Driver control keypad HT-011502 / 15 BUTTON
CAN CABLE +12V
CAN CABLE RETURN
10 / CAN, DASH, KEYPAD AND WAKE
M2 Electronic throttle body LS3 THROTTLE BODY
MOTOR A
MOTOR B
06 / DRIVE-BY-WIRE
J40 Haltech PD16 - connector A PD16 A / AMP34 KEY1
13 / HALTECH PD16 CONNECTIONS AND CONTROL
U23 Haltech TMS-4 receiver / HT-011600 HT-011600 / TMS-4 MCON6
16 / TMS-4 TPMS, ENGINE AND TRANSMISSION OIL TEMP
U24 WolfBox G900 Pro hardwire converter G900 PRO HARDWIRE KIT
U21 WolfBox G900 Pro mirror and front camera G900 PRO MIRROR / FRONT CAM
U25 WolfBox rear camera WOLFBOX REAR CAMERA
POWER + VIDEO CABLE
17 / WOLFBOX G900 PRO CAMERA POWER
GP009 Engine block / coil ground point
SP026 R3 CAN RETURN splice
SP032 COCKPIT 12V splice
SP070 Injector-only bank feed split
SP071 Coil bank feed split
LOOM101 Delta supplied return junction
SP116 HEADLIGHT POS splice
SP118 BRAKE POS splice
SP136 GT blue / blue permanent joint
OEM111-PG Stock odd coil PG splice
OEM113-PG Stock even coil PG splice
CUT-INJ-ODD-PWR Donor odd injector feed cut-end
CUT-INJ-EVEN-PWR Donor even injector feed cut-end
PROPOSED / physical location and hardware TBD
RETAINED OEM / actual location and construction to verify
EXPLICIT SPLICES / PROPOSED GROUND POINTS / SUPPLIER JOINS

PHYSICAL CONNECTION NOTES / NATIVE DESIGN HOLDS

All new joint/ground locations and material choices remain proposed. Electrical endpoint sets are checked against the native KiCad export.

PHYSICAL CONDUCTOR VIEW - READ FIRST

WHAT YOU BUILD: C / MAKE = connection you provide, using new wire or suitable reused donor wire/pigtail. S / LOOM = existing OEM/vendor loom wire; retain it. V / CABLE = intact supplied assembly to connect.

Read MAKE / LEAD / LOOM / CABLE tags, not ID prefixes, for construction work. LEAD = supplied harness wire needing end work; MAKE = additional connection/branch to provide. Inspect actual grouped leads before adding wire.

R3 HT-183200 and PD16 HT-188000 flying-lead harnesses are owner selected. Controller-side leads are supplied; free ends need routing/termination. Heavy main feeders and vehicle branches are separate. A drawn LEAD span can include an extension or adapter.

Delta loom, sensor branches, two pressure plugs, slip/OFF switches, warning lamp, fused feeds and IMU/private CAN are supplied assemblies. Do not fabricate duplicate branches or add sensor grounds. IMU cavities remain supplier-internal/unmapped.

Delta switch HOLD: current guide describes hardware solenoid-power interruption; historical pin35 ground-enable model is unresolved. C158 is HOLD, not a wire to make. Keep actual supplied switch loom intact pending revision verification.

Delta F41/F42/F43 represent supplied loom protection: current guide lists pump40A/solenoid20A/ignition10A; older pinout labels a25A supply. Confirm received loom before final coordination. Public CAN uses auxiliary port; private four-pin port is service only.

I / INTERNAL and B / BOND are not extra cut wires. U / HOLD is unresolved supplier construction. Wires_To_Provide.tsv and Supplied_Leads_To_Finish.tsv separate added wiring from lead end work; neither is a measured cut schedule.

Each solid C line is one individual proposed field conductor with exactly two ends. Bends and white crossing breaks do not add wire ends. W identifiers still name electrical nets.

S lines are individual supplied-loom conductor paths, retained within their supplier assembly; verify actual construction. V thick purple lines are intact multi-wire vendor cables, not single cut wires.

Short black bars inside SP nodes denote explicit proposed splices; inside TB1/TB2 they denote conducting metal. Repeated same pin labels share one actual terminal/stud; they are not additional cavities.

U3 is Cole Hersee 24059: large PWR from F5/2, large OUT to M1 small SOL; small CMD from R3 A18/HBO5 and small GND to GP129-C. These are installer tags, not stamped terminal numbers. Coil and contact supplies are separate.

D60 1N5408 is local across U3 small coil studs: band/K to CMD via C213; unbanded/A to GND via C214. C213/C214 are insulated diode leads, not long harness cuts. The W129 ground-stub depiction of C214 does not require another chassis landing. Verify polarity and prompt disengagement.

S2 clutch/neutral switch is removed by owner; C076/C123 and W046 retired. R3 C5/SPI1 is spare/unconnected, never jumpered. Actual start logic must omit that input requirement while retaining fresh request, engine-not-running, bounded crank and release controls.

Dashed I lines denote internal device/supplier continuity; dash-dot B lines denote metal/lug/bond continuity. Neither is an extra cut wire. Dotted U lines remain unresolved supplier paths.

Splice and ground-point positions, hardware, joining method, wire gauge/color, route, cut length and terminal parts are UNRELEASED. This diagram proposes connection topology, not measured vehicle geometry.

Ground points GP129-F/C/R are proposed front/control/rear vehicle-side returns. Their metal/bond network must be verified; no extra field wire is specified by dash-dot structure paths.

Chassis returns use short stubs and labelled ground symbols. Match W IDs; the graphic stub is not a measured cut length. GP129-F/C/R remain distinct proposed physical landing points on W129.

B005 GP129-F to GP129-C and B006 GP129-C to GP129-R are proposed chassis metal/bond continuity, represented by the W129 ground references, not extra drawn field wires.

W015 chassis/case bond references stay distinct from W129 return/strap terminals. No ground symbol is substituted for W002 isolated battery negative, W009 engine returns, sensor grounds, CAN returns or supplied Delta sensor returns.

J2/J3 ground straps and J21 ground assembly have two ends joined by their own assembly bodies. Those bodies are depicted inside their boxes, not as additional field wires.

CAN1 uses a proposed linear trunk R3 -> dash tap -> keypad tap -> TMS4 tap -> PD16. H/L pair IDs match each trunk/tap. J30 is the PD16 end-termination representation; no third resistor is added.

TB1 four branch wires and TB2 six branch wires (incoming directly from F0) land at explicit functional lug ports. Actual stud/stack assignments and block ratings/boots remain open. M1 B+ has two wires on the same battery-side stud.

Existing device pin/net assignments are retained. Stock L94 X111/X113 bank connector reference interfaces are added; functional device IDs do not imply inspected donor cavities.

S paths marked OEM-COIL or OEM-INJ are retained donor-loom conductors, not new branch wires. OEM splice nodes depict shared continuity; exact internal locations, wire sizes and condition require inspection.

Stock injector integration uses functional CUT-INJ boundaries, not an invented factory bulkhead plug. Isolate injector-only feed branches from factory coil/other loads; never bridge HCO1 and HCO2.

Stock coil low-reference grounding remains HOLD: retained C26 routing is not final approval; resolve GM separate low reference versus Haltech generic head-ground advice with actual coil/R3 evidence.

Use Physical_Connections.tsv for ID/from/to/kind/pair and blank measurement fields, Physical_Junctions.tsv for joint intent, and Physical_Topology.json as the maintained topology source. Never overwrite measured entries by re-initializing it.

POWER-SYSTEM SCOPE / RELEASE HOLDS

All 43 selected power/control/vendor-link nets are complete. Other component pins, sensor networks, CAN data and driver signals are omitted, not declared NC. See full continuous sheet for those conductors.

GT M6_IGN -> S1 -> R3 A13 only. K1 removed by owner; F0 feeds TB2 directly on W003 MAIN_FUSED_BUS. Verify engine-stop, PD16 OFF/CAN-loss and GT timing; positive feeders remain connected.

S40 is the single owner-ordered BAOMAIN latching NC stop, actual contact markings to verify. Blue/blue flying leads join at SP136; no external switch or chassis connection.

TB1 is proposed Sierra FS46540 common M10 post. TB2 is proposed Summit SUM-900533; specific250A vs family150A listing conflict remains open. Verify allowed stacks, torque, covers and hot continuous duty.

F2 shared trunk near battery -> M1 battery B+; separate wire same stud -> F3 near starter -> G1. F2/F3 retained, no removal authorized. Actual feeder fuse values/cable sizing and both-source fault checks remain open.

PD16 A31-A34 each has an individual ground conductor. R3 A11 CAN return stays separate from vehicle grounds. Injector PWR branches return via omitted driver wiring; coil power grounds are included.

WolfBox BATT/ACC are separate wires sharing one explicit switched-feed splice. Preserve regulated5V USB-C and rear-camera V assemblies. GPS/reverse lead unused; verify received kit and switched startup.

R3/PD16 channel assignments, separate lift pumps, full-speed fan control and no injector/coil/lift bank fuses are unchanged. No actual NSP tune or powered testing is claimed.

REV J / PHYSICAL CONDUCTOR VIEW 01 | 70 native component interfaces | 43 native nets | C85 / S41 / V2 / I8 / B9 / U0

Connection topology is proposed and geometry is illustrative. No measured cut lengths, fabrication release, installed NSP settings or powered verification claimed.